

Learning Tactile-Aware Quadrupedal Loco-Manipulation Policies

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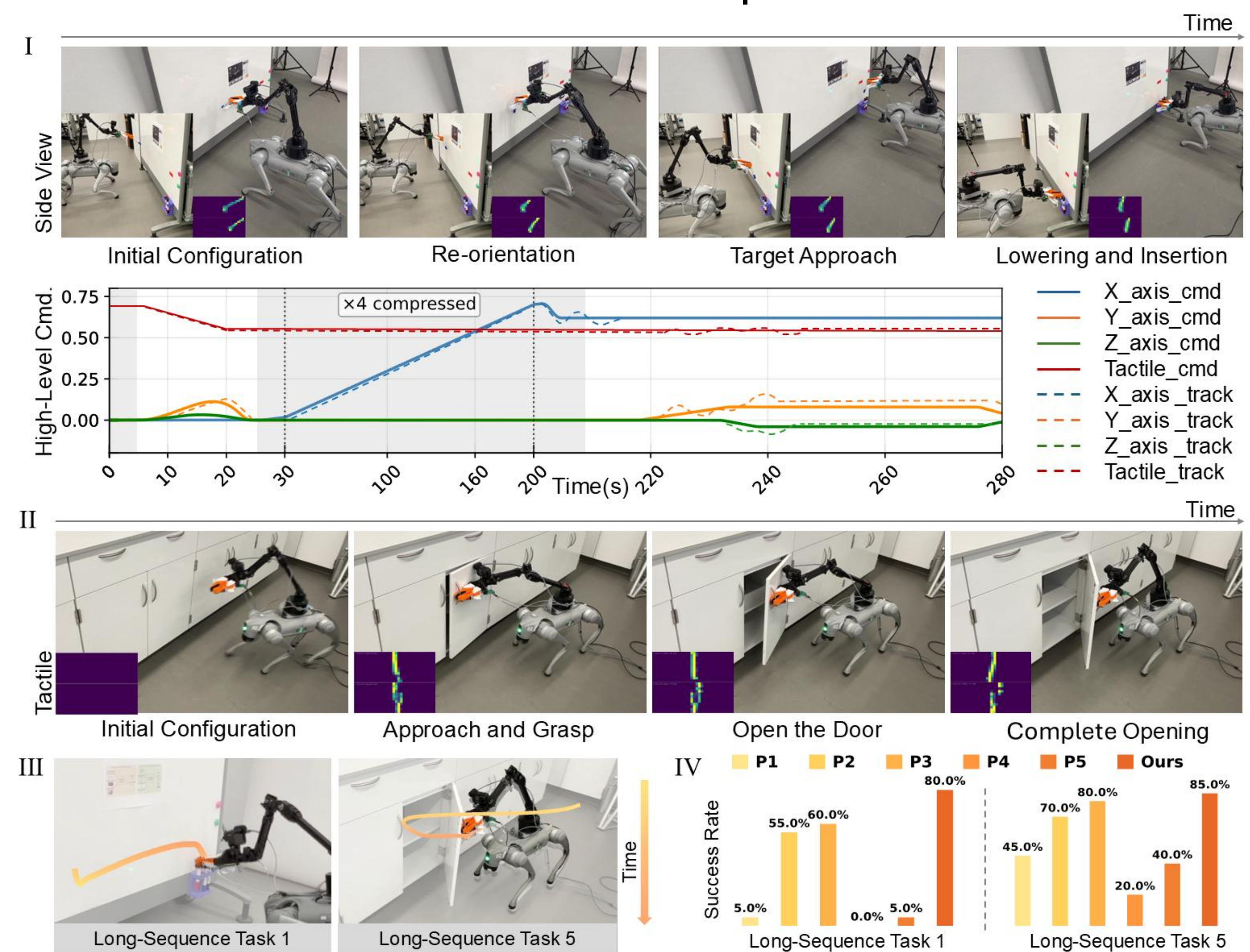
website

Overview

Tactile-aware hierarchical whole-body control policies for quadrupedal loco-manipulation.

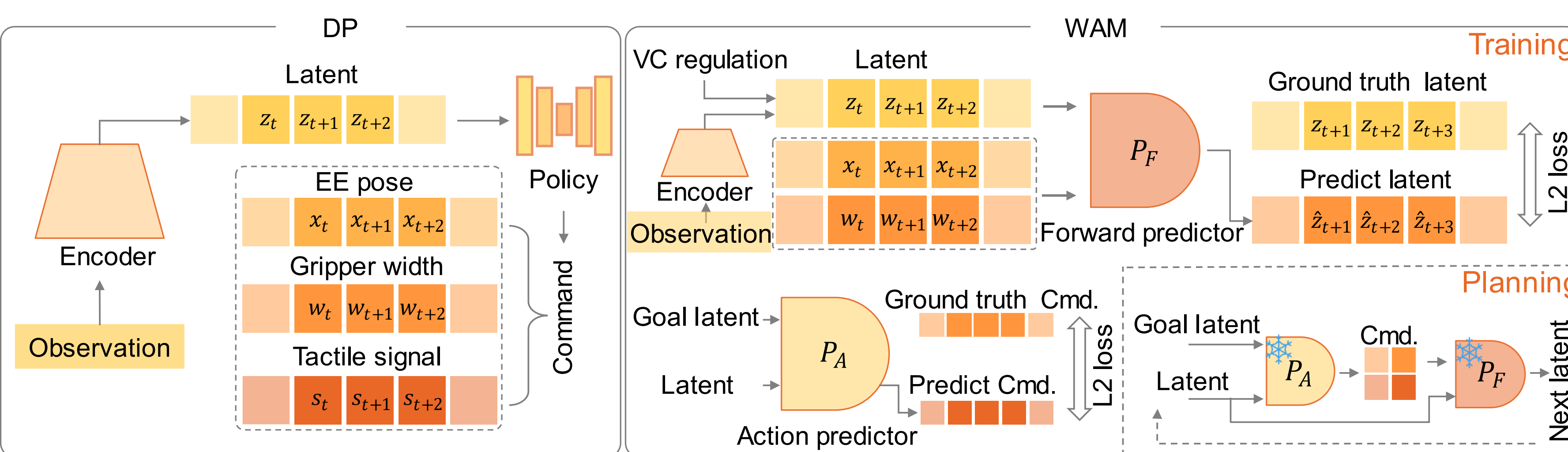
Loco-Manipulation Exp.

Challenging long-horizon, complex, and contact-rich loco-manipulation tasks.



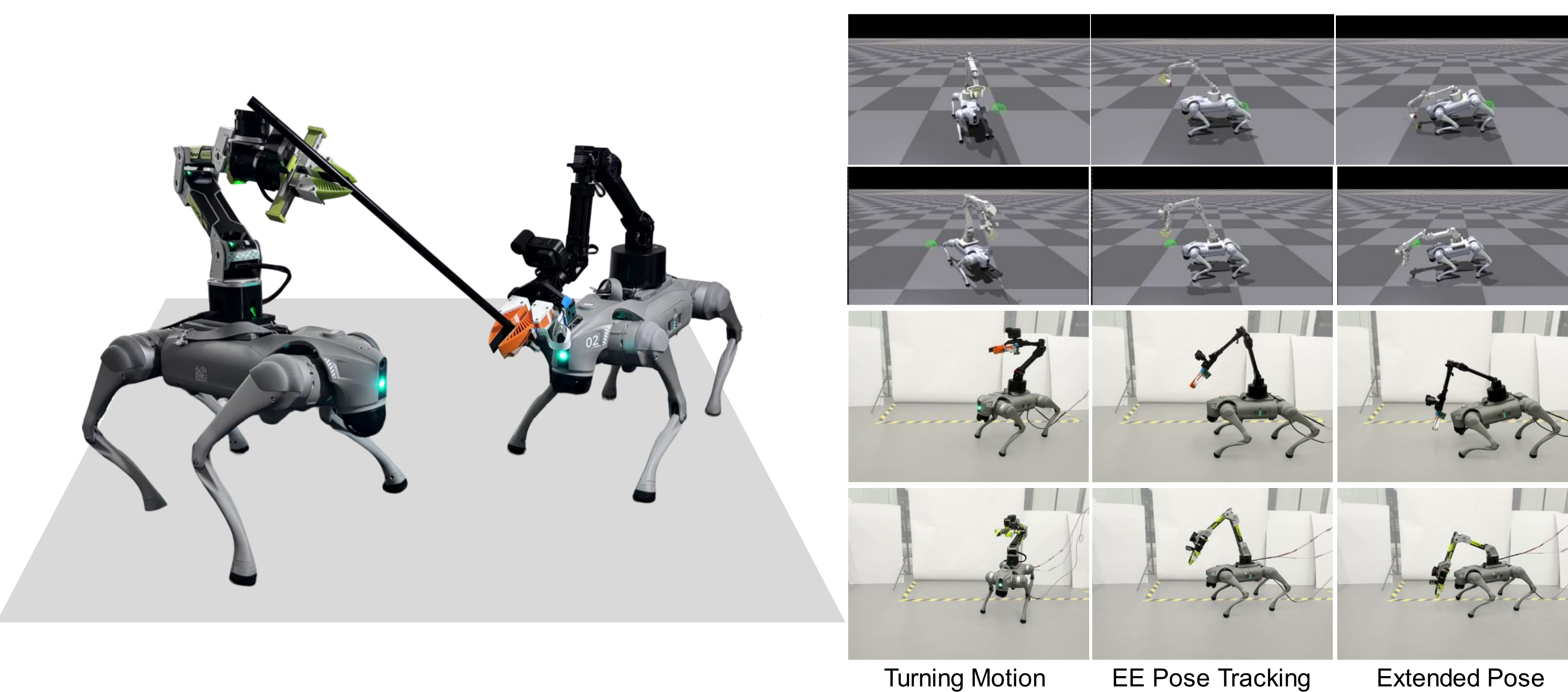
Framework

Diffusion policy based and world action model based high-level policy.



Cross Manipulator Whole-Body Control

Across substantially different manipulator configurations, demonstrating the effectiveness and generalizability.



Real-Word Benchmark

Evaluating the contribution of tactile.

	Task1 Re-ori	Task1 Insertion	Task1 Whole	Task2	Task3	Task4	Task5
P1	0.35	0.15	0.05	0.35	0.20	0.35	0.45
P2	0.95	0.70	0.55	0.80	1.00	0.85	0.70
P3	1.00	0.70	0.60	0.80	1.00	0.95	0.80
P4	0.20	0.30	0.00	0.10	0.10	0.10	0.20
P5	0.40	0.35	0.05	0.10	0.15	0.45	0.40
Ours	1.00	0.85	0.80	0.85	1.00	0.95	0.85

